Investigation: CIA SANEAMENTO DO PARANA - SANEPAR (CVM code 18627), fiscal year 2023, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Saneamento, Serv. Água e Gás (origin: FCA); group: non-financial.
Trading codes (2023 FCA, version 1, filed on 2023-05-02): SAPR11 (Units, Bolsa, since 2017-11-22); SAPR3 (Ações Ordinárias, Bolsa, since 2016-12-21); SAPR4 (Ações Preferenciais, Bolsa, since 2016-12-21).

Document used

2023 DFP, version 1, filed on 2024-02-08; origin: open data; type: individual; original scale: MIL.
Versions known on the date: 1 (2024-02-08).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=133929

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −2.556 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 48 (n = 26)
- Percentile in group and year: 57 (n = 530)
- The company's own history: median −2.599; Tukey range (1.5 × IQR): from −2.958 to −2.098; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 1.046 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 84 (n = 29)
- Percentile in group and year: 61 (n = 599)
- The company's own history: median 1.026; Tukey range (1.5 × IQR): from 0.775 to 1.322; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

GMI, gross margin index
- Value: 0.966 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 50 (n = 29)
- Percentile in group and year: 41 (n = 609)
- The company's own history: median 1.002; Tukey range (1.5 × IQR): from 0.958 to 1.043; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.003 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 53 (n = 29)
- Percentile in group and year: 53 (n = 610)
- The company's own history: median 0.990; Tukey range (1.5 × IQR): from 0.989 to 0.992; value above the range (index; 5 fiscal years)
- Confidence in the data: medium

SGI, sales growth index
Value: 1.109 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 19 (n = 29)
Percentile in group and year: 56 (n = 610)
The company's own history: median 1.084; Tukey range (1.5 × IQR): from 1.054 to 1.112; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DEPI, depreciation index
Value: 0.964 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 33 (n = 26)
Percentile in group and year: 53 (n = 549)
The company's own history: median 1.035; Tukey range (1.5 × IQR): from 0.908 to 1.156; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.824 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 36 (n = 29)
Percentile in group and year: 18 (n = 606)
The company's own history: median 0.916; Tukey range (1.5 × IQR): from 0.577 to 1.415; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.047 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 58 (n = 30)
Percentile in group and year: 56 (n = 628)
The company's own history: median −0.044; Tukey range (1.5 × IQR): from −0.060 to −0.021; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

LVGI, leverage index
Value: 1.021 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 60 (n = 29)
Percentile in group and year: 59 (n = 619)
The company's own history: median 1.006; Tukey range (1.5 × IQR): from 0.966 to 1.071; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DPO jump
Value: 6.0 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 25.7 days)
Percentile in sector and year: 74 (n = 29)
Percentile in group and year: 70 (n = 591)
The company's own history: median −0.9 days; Tukey range (1.5 × IQR): from −4.5 days to 3.8 days; value above the range (5 fiscal years)
Confidence in the data: medium

Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: −1.82 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 22 (n = 30)
Percentile in group and year: 23 (n = 588)
The company's own history: median 3.18 p.p.; Tukey range (1.5 × IQR): from −1.45 p.p. to 8.99 p.p.; value below the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.050 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 58 (n = 30)
Percentile in group and year: 56 (n = 627)
The company's own history: median −0.045; Tukey range (1.5 × IQR): from −0.063 to −0.023; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 2.858 (index)
Components: X1 = 0.043; X2 = 0.187; X3 = 0.125; X4 = 1.076 (indices)
Percentile in sector and year: 76 (n = 31)
The company's own history: median 2.881; Tukey range (1.5 × IQR): from 2.400 to 3.534; value inside the range (index; 5 fiscal years)
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 7/9 (9 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA (ΔROA): 1
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 1
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): 1
- Change in gross margin (ΔMARGIN): 1
- Change in asset turnover (ΔTURN): 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- DFP 2018, v1: 2019-02-05
- DFP 2018, v2: 2019-02-21
- DFP 2019, v1: 2020-02-06
- DFP 2020, v1: 2021-02-11
- DFP 2021, v1: 2022-02-24
- DFP 2022, v1: 2023-02-16
- FCA 2023, v1: 2023-05-02
- DFP 2023, v1: 2024-02-08

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).

Excerpt checked literally: "Em nossa opinião, as demonstrações contábeis acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da Companhia em 31 de dezembro de 2023, o desempenho de suas operações e os seus fluxos de caixa para o exercício findo nessa data, de acordo com as práticas contábeis adotadas no Brasil e as normas internacionais de relatório financeiro (IFRS) emitidas pelo International Accounting Standards Board (IASB)."

Auditor in the 2021 FCA: BDO RCS Auditores Independentes SS.

Auditor in the 2022 FCA: BDO RCS Auditores Independentes SS.

Auditor in the 2023 FCA: BDO RCS Auditores Independentes SS.

Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2012: entry of KPMG Auditores Independentes, with no other firm leaving.
- 2012: BDO Trevisan Auditores Independentes -> Ernst & Young Terco Auditores Independentes S.S.. The outgoing firm audited 1 consecutive fiscal years (2011 to 2011), by the start and end dates in the FCA.
- 2013: exit of KPMG Auditores Independentes, with no other firm entering. The FCA does not have the dates needed to count the outgoing firm's fiscal years.
- 2017: Ernst & Young Terco Auditores Independentes S.S. -> BDO RCS Auditores Independentes SS. The outgoing firm audited 5 consecutive fiscal years (2012 to 2016), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 39 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:41:47+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.